Project 3: Potentiometer Controlled Robotic Arm Using Arduino Uno

CSAI 205: Fundamentals of Circuits and Electronics University of Science and Technology, Zewail City Instructor: Assoc. Prof. Mohamed Maher Ata

Name: **Hamza Moamed**

1. Objective

The goal of this project is to build a robotic arm with three joints, where each joint is moved by a servo motor and controlled by its own potentiometer. The Arduino Uno reads the position of each potentiometer and sets the matching servo to the same relative angle, so turning a knob moves the corresponding joint of the arm.

The circuit was first simulated and then built, together with the arm structure, as a working prototype.

2. Components Used

Component	Quantity
Arduino Uno	1
Servo motors	3
Potentiometers 50 kohm	3
Breadboard	1
Jumper wires (male-male and male-female)	about 20
External 5 V supply	1
Arm structure: **[plastic]**	1

3. Background

Potentiometer as a voltage divider. A potentiometer has three terminals. The two outer terminals are connected to 5 V and GND, and the middle terminal (the wiper) gives a voltage between 0 V and 5 V depending on the knob position. With a 50 kohm potentiometer the current drawn is only 5 V / 50 kohm = 0.1 mA, so it can safely be powered from the Arduino.

Analog input. The Arduino has a 10-bit analog-to-digital converter. The function analogRead gives a number from 0 to 1023 for an input from 0 V to 5 V, so each step is about 5 V / 1023 = 4.9 mV.

Mapping to an angle. The function map converts the reading (0 to 1023) into an angle (0 to 180 degrees), so one step of the reading is about 0.18 degrees.

Servo motor. A servo has a motor, a gearbox and position feedback inside. It is controlled by a pulse signal repeated about every 20 ms. The pulse width (about 1 ms to 2 ms) sets the angle. The Servo library of the Arduino generates this signal, and the function write(angle) sets the position.

External power supply. A servo can draw a large current while it moves, much more than an Arduino pin or the 5 V pin of the board can supply. Therefore the three servos are powered from an external 5 V source, as required by the project. The Arduino 5 V pin is not used for the servos.

Common ground. The ground of the external supply, the ground of the servos and the Arduino GND must be connected together. The control signal is measured relative to ground, so without a common reference the servos would not respond correctly.

4. Circuit Design

Part	Connection
Potentiometer 1 (middle terminal)	A0
Potentiometer 2 (middle terminal)	A1
Potentiometer 3 (middle terminal)	A2
Servo 1 signal	pin 4
Servo 2 signal	pin 3
Servo 3 signal	pin 2
Potentiometers, outer terminals	5 V and GND of the Arduino
Servos, power wire	+ 5 V of the external supply
Servos, ground wire	common GND
External supply GND	Arduino GND

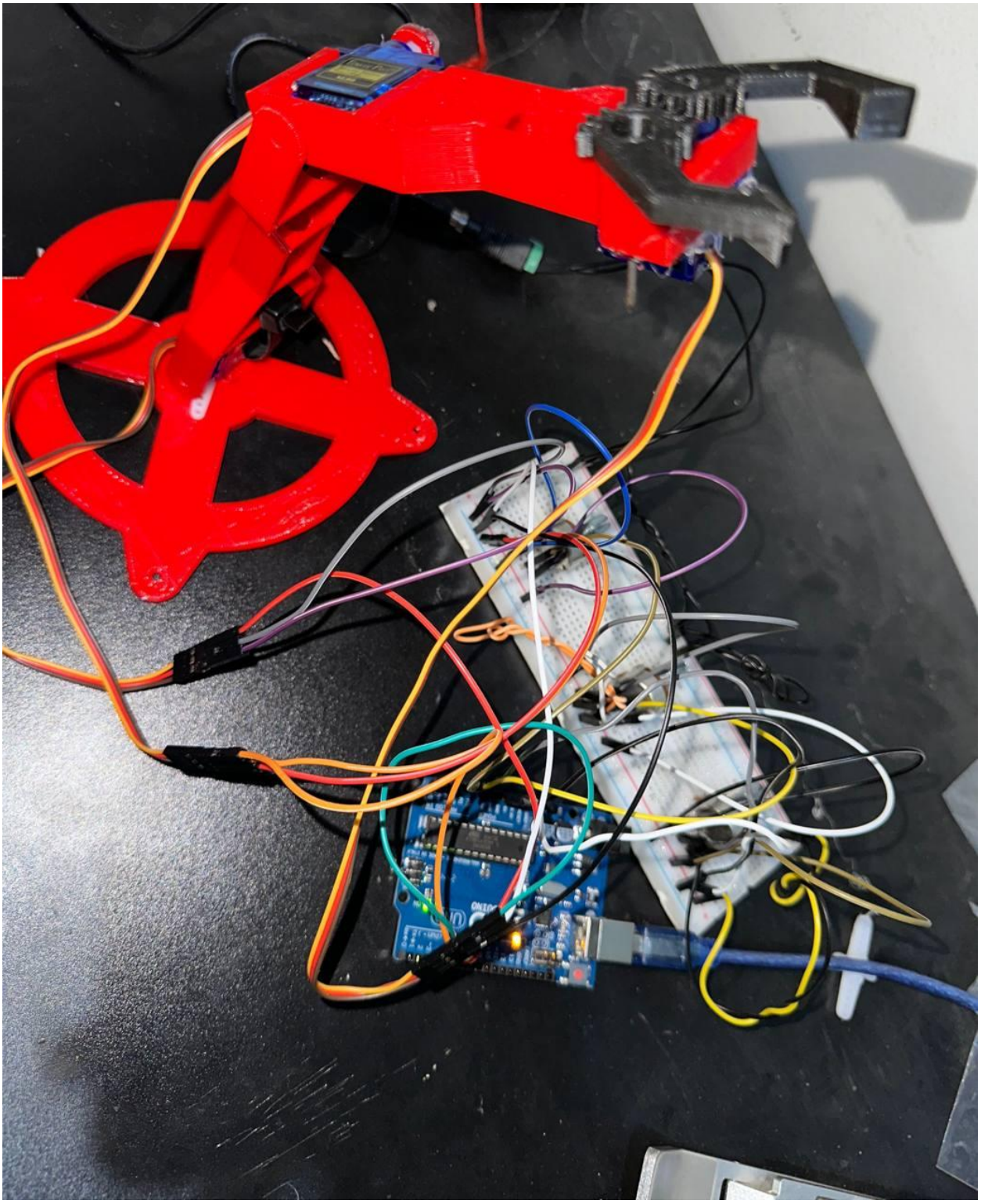
6. Simulation

Potentiometer position	Expected servo angle	Observed
Minimum	about 0 degrees	**[result]**
Middle	about 90 degrees	**[result]**
Maximum	about 180 degrees	**[result]**

In the simulation the external supply showed 5.00 V and a current of about 10.6 mA with the servos at rest.

7. Hardware Implementation

The potentiometers, Arduino and breadboard were connected as in the circuit design. The three servos were powered from the external 5 V source and their ground was connected to the Arduino GND. The program was uploaded through the USB cable using the Arduino IDE.



8. Problems Faced and Solutions

Examples of things that often happen: servos shaking or the Arduino resetting when they move, a missing common ground, a servo connected to the wrong pin, a joint hitting its mechanical limit, or a potentiometer wired with its terminals swapped so the direction is reversed.

9. Conclusion

We built a potentiometer controlled robotic arm with three servo joints driven by an Arduino Uno. The Arduino reads each potentiometer, converts the reading to an angle and moves the matching servo. The project showed how an analog input is converted and used to control a motor, why servos need a separate supply with a common ground, and how a simple program can control a mechanical system.